POWER SYSTEM AND COMMUNICATION PROJECT

PROJECT TITLE : Footstep-Based Power Generation with Integrated Crowd Density Monitoring and Stampede Alert System



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INTRODUCTION

Rapid urbanization and population growth have resulted in extremely high crowd densities in public places such as railway stations, metro platforms, religious gathering sites, stadiums, shopping malls, and public events. While such crowds indicate active public participation, they also introduce serious safety hazards, particularly stampede incidents. Stampedes are sudden, uncontrolled crowd movements that often occur due to panic, bottlenecks, restricted exits, or sudden surges in crowd inflow. These incidents can lead to severe injuries, loss of life, and large-scale chaos.

At the same time, crowded environments continuously dissipate a significant amount of mechanical energy in the form of human footsteps. This energy, although small per step, becomes substantial when accumulated over thousands of people. Traditionally, this energy is wasted without any useful utilization. The proposed project, titled “Footstep-Based Power Generation with Integrated Crowd Density Monitoring and Stampede Alert System,” aims to address both these challenges simultaneously by converting wasted mechanical energy into electrical power and using that power to monitor crowd behavior and enhance public safety.

This project integrates renewable energy harvesting, embedded systems, and IoT technology into a single smart system. The system not only generates electrical energy from footsteps using piezoelectric sensors but can also continuously monitor crowd density and movement patterns to detect early signs of stampede risk and issue timely alerts.

OBJECTIVE

The primary objective of this project is to design and implement a self-powered intelligent system that enhances public safety while promoting sustainable energy usage. The specific objectives are:

- To generate usable low-voltage DC electrical power from human footsteps using piezoelectric sensors.
- To efficiently rectify and regulate the harvested energy for powering electronic components.
- To continuously monitor crowd movement and estimate crowd density in real time.
- To detect abnormal crowd behavior and early warning signs of stampede conditions using multiple decision parameters.
- To provide immediate alerts through local indicators and IoT-based mobile notifications.
- To demonstrate an integrated power–safety system that remains operational even during power outages or emergency situations.

MOTIVATION AND IDEA DEVELOPMENT

The motivation for this project originates from two major real-world observations. First, stampede incidents continue to occur in crowded public places, especially in developing countries, due to inadequate monitoring systems and delayed response mechanisms. Many existing safety systems depend heavily on external power supply and human supervision, which makes them unreliable during emergencies such as power failures or disasters.

Second, there is a growing need for renewable and sustainable energy solutions. Human movement is a readily available and renewable source of energy in public spaces, yet it is largely untapped. Piezoelectric materials offer a practical method to convert mechanical stress into electrical energy, making them suitable for footstep-based power generation.

The idea development involved combining these two independent concepts—energy harvesting and crowd safety—into a unified system. Instead of treating power generation and safety monitoring as separate demonstrations, this project integrates them such that the harvested energy directly supports the operation of the safety system. This approach improves system reliability, reduces dependence on grid power, and enhances relevance in real-world scenarios.

COMPONENTS USED

The project consists of the following major hardware and software components:

- **Piezoelectric Sensors:** These sensors generate electrical voltage when mechanical stress is applied due to footsteps. They serve as the primary energy source for the system.
- **Bridge Rectifier:** Since the output from piezoelectric sensors is alternating in nature, a bridge rectifier is used to convert AC voltage into DC voltage.
- **Filter Capacitors:** Capacitors are used to smooth out voltage fluctuations after rectification and provide stable DC output.
- **Rechargeable Battery:** A rechargeable battery (power bank) is used to supply stable power to the ESP32, allowing the system to operate continuously even when footstep energy generation is low or temporarily unavailable.
- **ESP32 Microcontroller:** The ESP32 acts as the central control unit. It processes sensor data, executes decision logic, manages power usage, and handles IoT communication through built-in Wi-Fi.
- **Resistors:** The resistors are used to limit the current flow, thereby protecting the LED and the ESP32 from damage.
- **LED:** The LED is used to provide a visual indication of system status or alert conditions. It indicates whether the system is powered ON using energy generated from footsteps or stored energy. This confirms that the power generation and storage unit is working properly.
- **IoT Platform (Blynk):** Used for real-time data visualization, remote monitoring, and push notification alerts.

WORKING LOGIC OF THE SYSTEM

Footstep Detection Logic

The system uses a peak-based detection method to identify individual footsteps. When a person steps on the piezoelectric sensor, a voltage spike is generated due to mechanical pressure. If this voltage rises above a predefined threshold value, the system recognizes it as a valid footstep.

To prevent multiple counts from a single step (caused by vibrations or sensor oscillations), a lockout time is applied. During this short lockout period, any additional voltage peaks are ignored. This ensures that each physical footstep is counted only once, improving accuracy.

Crowd Alert Logic

The detected footsteps are continuously counted within a five-second time window. The system analyzes how many steps occur during this short duration rather than only counting total steps.

If more than five footsteps are detected within the five-second window, it indicates unusually rapid movement or crowd buildup. In such cases, the system triggers a crowd alert, signaling possible abnormal crowd behavior.

IoT Monitoring using Blynk

The ESP32 connects to the Blynk cloud platform using its built-in Wi-Fi module. Sensor data such as footstep count, raw piezoelectric sensor values, and crowd alert status are transmitted continuously to the cloud. The Blynk mobile application displays this information in real time, allowing users to monitor crowd conditions remotely.

When abnormal crowd movement is detected and an alert is triggered, the system sends push notifications through the Blynk application. This enables authorities or event organizers to receive instant warnings and take timely preventive action.

WORKING PROCESS OF THE SYSTEM

The working of the proposed system can be clearly understood by dividing it into the three units defined in the initial concept:

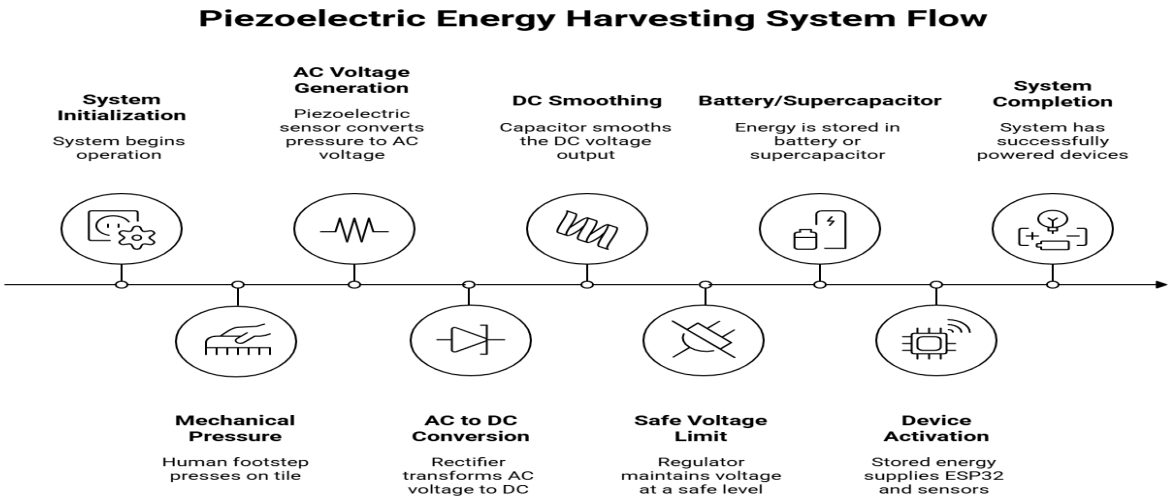
1. Footstep Power Generation Unit

2. Crowd Monitoring & Stampede Detection Unit

3. IoT Alert & Visualization Unit

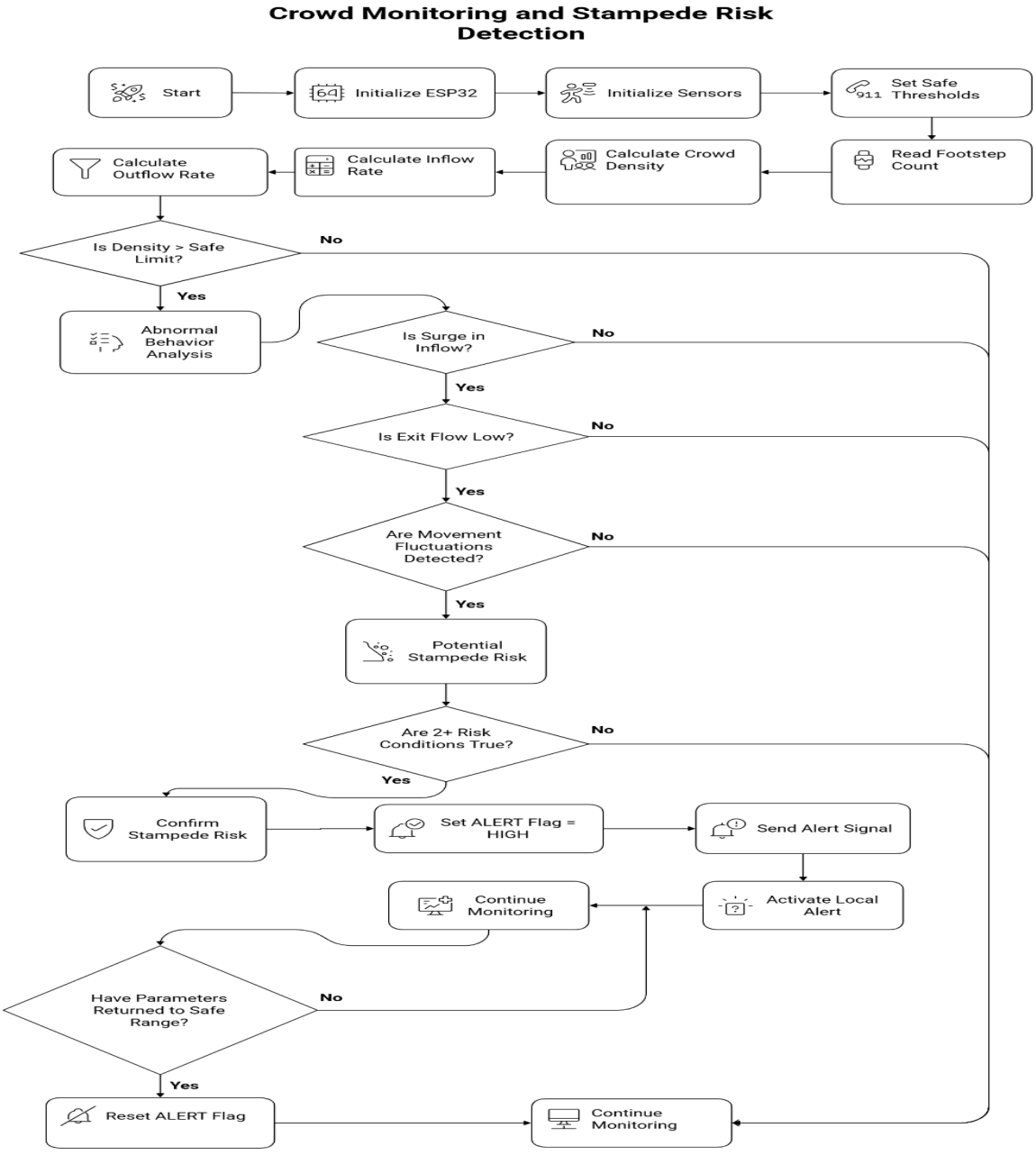
Each unit operates independently but is tightly coordinated by the ESP32 microcontroller. The complete system operation follows a sequential and event-driven flow, as explained below using textual flowcharts.

Unit 1 : Footstep Power Generation Unit



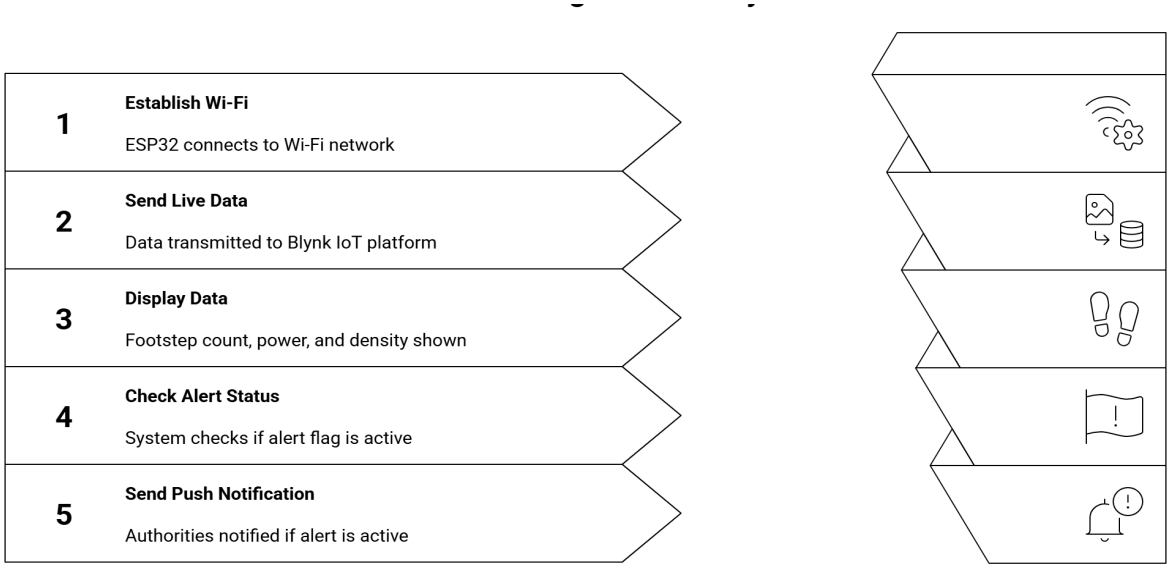
Whenever a person walks over the piezoelectric tile, mechanical stress is generated. This stress produces an alternating electrical voltage due to the piezoelectric effect. Since electronic systems require DC power, the AC voltage is rectified, filtered, and regulated. The processed electrical energy is then stored and later used to power the ESP32 and sensing units, making the system self-sustaining.

Unit 2: Crowd Monitoring & Stampede Detection Unit



The ESP32 continuously processes real-time data from footstep and crowd sensors. Instead of triggering alerts based on crowd size alone, the system evaluates multiple parameters such as crowd density, rate of change, and movement irregularities. Only when several abnormal conditions occur together does the system identify a stampede risk, thereby reducing false alarms.

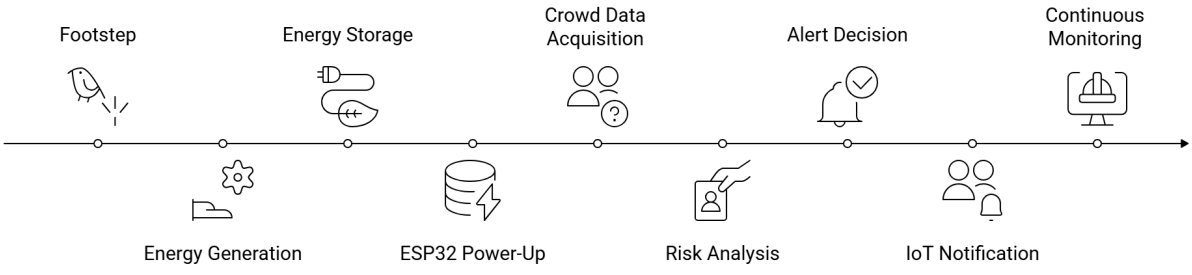
Unit 3: IoT Alert & Visualization Unit



Once the ESP32 detects a stampede risk, it immediately communicates the alert status to the IoT platform using Wi-Fi. The Blynk application displays live system data and sends push notifications to responsible personnel. This enables rapid decision-making and early intervention before panic escalates.

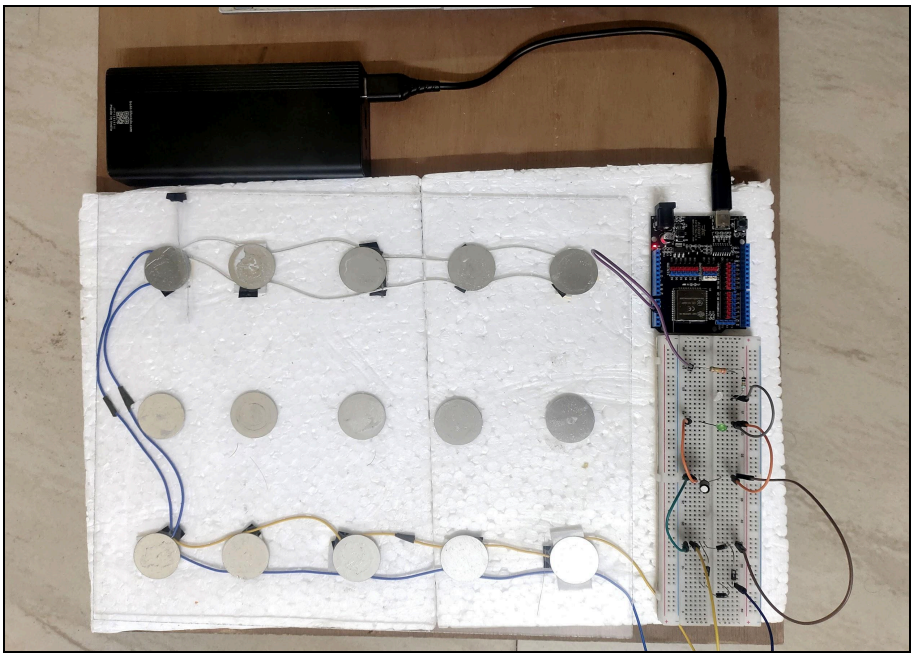
Overall Integrated System Flow

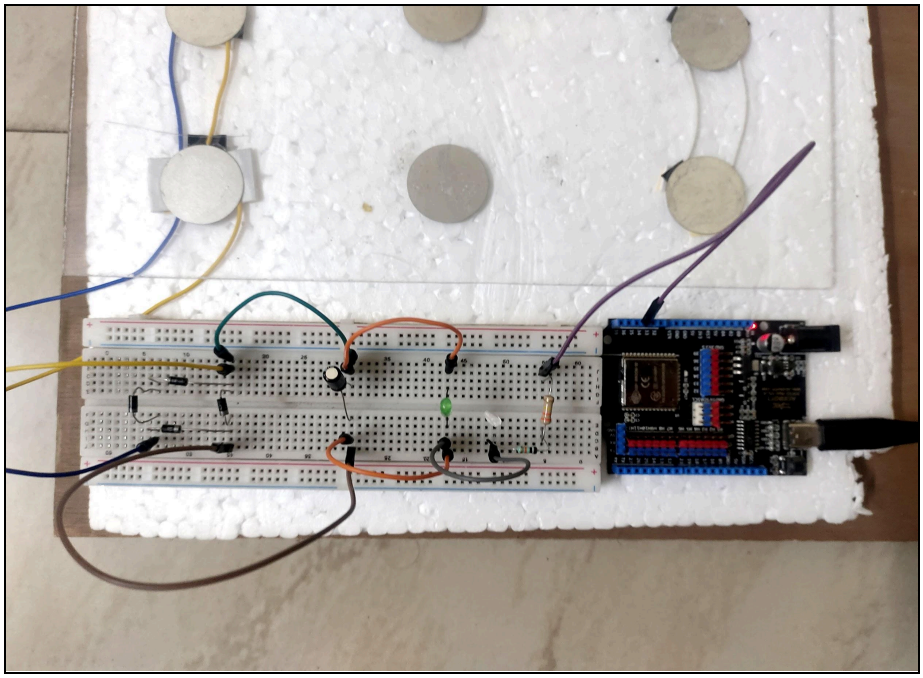
The complete system follows this high-level flow:

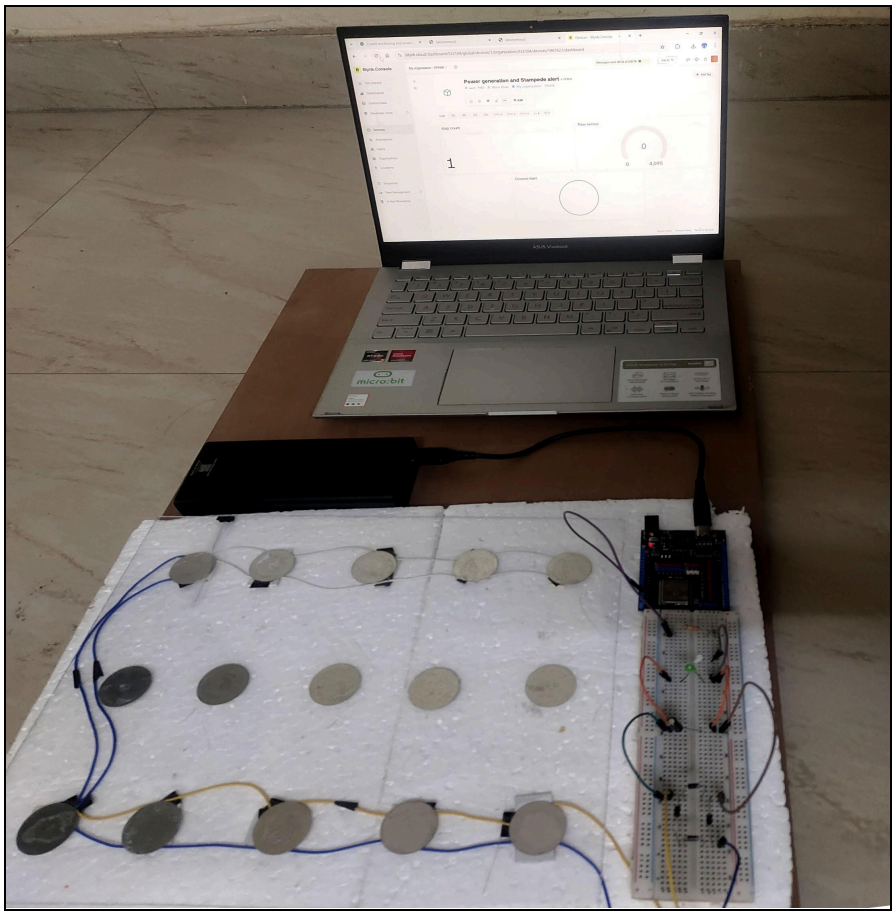


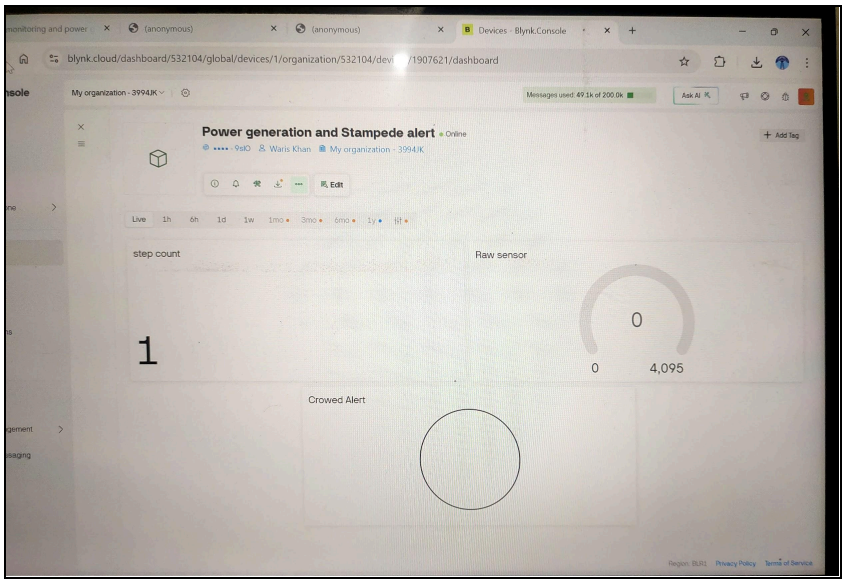
This structured flow ensures that energy harvesting and safety monitoring operate as a single unified system, making the solution reliable, sustainable, and suitable for real-world deployment.

HARDWARE IMPLEMENTATION (EXPERIMENTAL SETUP)






micro:bit


Devices - Blynk.Console
blynk.cloud/dashboard/532104/global/devices/1/organization/532104/dev /1907621/dashboard
My organization - 3994JK
Messages used: 49.1k of 200.0k
Ask AI
Power generation and Stampede alert
Online
Waris Khan
My organization - 3994JK
Add Tag
Edit
Live
1h
6h
1d
1w
1mo
3mo
6mo
1y
step count
1
Raw sensor
0
0
4,095
Crowed Alert
Region: BLR1
Privacy Policy
Terms of Service

SOFTWARE IMPLEMENTATION (ARDUINO CODE)

```
#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID   "TMPL3OjDoD2y2"
#define BLYNK_TEMPLATE_NAME "Stampede alert"
#define BLYNK_AUTH_TOKEN    "FJbSnOwTAyUonyHCzcwrl7QP4AGC9sIO"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

// WiFi credentials
const char* ssid = "1";
const char* password = "11223344";

// Piezo configuration
const int piezoPin = 34;
const int STEP_THRESHOLD = 2400;
const unsigned long STEP_LOCKOUT = 350;
unsigned long lastStepTime = 0;
int stepCount = 0;

// Step window logic
unsigned long windowStartTime = 0;
int stepsInWindow = 0;

// Alert parameters
const int STEP_WINDOW_THRESHOLD = 5;
```

```
const unsigned long WINDOW_DURATION = 5000;
bool alertSent = false;

void setup()
{
 Serial.begin(115200);
 WiFi.begin(ssid, password);
 while (WiFi.status() != WL_CONNECTED) {
 delay(500);
}
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, password);
  windowStartTime = millis();
}

void loop()
{
  Blynk.run();

  int piezoValue = analogRead(piezoPin);
  unsigned long now = millis();

  Blynk.virtualWrite(V3, piezoValue);

  if (piezoValue > STEP_THRESHOLD &&
      (now - lastStepTime) > STEP_LOCKOUT)
{
    stepCount++;
```

```
    stepsInWindow++;
    lastStepTime = now;

    Blynk.virtualWrite(V1, stepCount);

    while (analogRead(piezoPin) > STEP_THRESHOLD) {
      delay(2);
    }
  }
  if (now - windowStartTime >= WINDOW_DURATION)
{
    if (stepsInWindow > STEP_WINDOW_THRESHOLD && !alertSent)
{
      Blynk.virtualWrite(V4, 1);
      Blynk.logEvent( "crowd_alert", "⚠️ Crowd surge detected: >5 steps in 5 seconds");
      alertSent = true;
    }

    if (stepsInWindow <= STEP_WINDOW_THRESHOLD) {
      Blynk.virtualWrite(V4, 0);
      alertSent = false;
    }

    stepsInWindow = 0;
    windowStartTime = now;
  }
}
```

ADVANTAGES

- The proposed system utilizes piezoelectric energy harvesting, which makes use of mechanical energy generated from human movement. This approach is environmentally sustainable and reduces dependence on conventional power sources.
- The system provides real-time crowd monitoring by continuously analyzing footstep activity. This enables early identification of abnormal crowd movement, allowing preventive action to be taken before critical conditions develop.
- The hardware architecture is cost-effective and modular, making it suitable for deployment in both small and large public spaces. Additional sensing units can be integrated without major changes to the core system design.
- Due to its integrated alert mechanism and autonomous operation, the system is well suited for public safety and crowd management applications, particularly in high-footfall environments.

LIMITATIONS

- The electrical power generated through footstep-based piezoelectric sensors is relatively low in magnitude, limiting its application to low-power electronic devices such as microcontrollers and sensors.
- The system requires environment-specific calibration, as factors such as flooring material, crowd behavior, and footstep force can affect sensor output and detection accuracy.
- System performance is also influenced by sensor placement and mechanical coupling. Improper installation may result in reduced sensitivity or inaccurate footstep detection.

APPLICATIONS

- The proposed system is suitable for various high-footfall and safety-critical environments, including railway stations and metro platforms where crowd congestion is frequent.
- It can be deployed at religious gathering places such as temples, mosques, and churches during festivals and special events.
- The system is applicable in stadiums, concert venues, and exhibition centers to prevent crowd-related accidents.
- It can be integrated into smart city infrastructure for sustainable energy utilization and public safety monitoring.
- The system is also useful in disaster-prone areas where reliable power supply and early warning systems are essential.

FUTURE SCOPE

1. **Distributed Sensing Architecture:** The system can be extended by deploying multiple piezoelectric sensor nodes across a larger area. This would enable more accurate spatial analysis of crowd density and movement patterns, improving reliability in large venues.
2. **Advanced Data Processing and Analytics:** Integration with cloud platforms can allow long-term storage and analysis of crowd data. This can support trend analysis, peak-hour prediction, and improvement of alert thresholds based on historical data.
3. **Alternative Communication Technologies:** To overcome limitations of Wi-Fi coverage, the system can be enhanced using GSM or LoRa communication modules. This would enable long-range and low-power data transmission, especially in remote or large-scale public areas.
4. **Enhanced Energy Management:** Improved power conditioning and energy storage techniques can be implemented to increase efficiency. The harvested energy can be utilized to power additional low-power IoT devices, reducing dependency on external power sources.

5. **Integration with Safety Infrastructure:** The system can be interfaced with public address systems, digital display boards, or emergency lighting to provide automated crowd control guidance during high-risk situations.

CONCLUSION

The Footstep-Based Power Generation with Integrated Crowd Density Monitoring and Stampede Alert System presents an innovative and practical solution to two critical urban challenges: energy wastage and public safety. By harvesting renewable energy from human footsteps and using it to power an intelligent crowd monitoring system, the project demonstrates an efficient, self-sustaining, and reliable approach to safety management.

The integration of piezoelectric energy harvesting, ESP32-based embedded control, and IoT communication ensures continuous monitoring and timely alerts, even during power outages. This project highlights the potential of combining renewable energy technologies with smart safety systems, making it highly relevant for modern public infrastructure and future smart city applications.

REFERENCES

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- N. K. Suryadevara and S. C. Mukhopadhyay, "*Wireless sensor network based smart monitoring of public environments,*" IEEE Sensors Journal, vol. 15.